

Weighing the Coin: Logit-Level Confabulation Correction at the Geometric Decision Point

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Abstract

Language models encode uncertainty about their own knowledge that far exceeds what they express: a model fabricating a confident answer about a fictional entity nonetheless shows nonzero entropy at the critical generation token (token 30, entropy = 0.24 at baseline). We report that a constant logit bias toward hedge tokens produces a sharp phase transition from confident fabrication to appropriate epistemic hedging on ablated models. In a powered study (175 trials, Qwen3.5-27B, 20 fictional entities), the uncertainty signal at the decision token increases monotonically with bias strength ($0.24 \rightarrow 0.77$, a $3.2\times$ amplification at bias = 5.0), and the confabulation rate drops from 77% to 51%. The intervention is selective: legitimate Fermi estimation is unperturbed at all bias levels. We identify two mechanistically distinct confabulation subtypes—fabrication (inventing entities, requiring cache-level intervention) and imprecision (failing to hedge, responding to logit-level correction)—and show that the dose required for correction correlates with the strength of the model’s knowledge anchoring. Because the intervention operates on the output distribution rather than the representation space, it sidesteps the detection–correction gap documented by Basu et al. [1]: probes detect failure modes at 98.2% AUROC but representation-level steering corrects only $\sim 20\%$ while disrupting 53%. Logit-level correction avoids this collision entirely.

1 Introduction

Language models know more than they say. A model asked about the depth of the “Grenvillia Trench”—a fictional entity—generates a confident, fabricated answer complete with invented measurements. Yet the model’s internal state tells a different story: at the critical generation token (approximately token 30, where the model commits to fabrication or hedging), the entropy is 0.24—low, but nonzero. The uncertainty is present. It is not expressed.

This knowledge–expression gap is a specific instance of a broader field-wide finding. Basu et al. [1] demonstrate that linear probes discriminate failure modes at 98.2% AUROC while model output sensitivity is only 45.1%—a 53-percentage-point knowledge–action gap. Attempts to close the gap via representation-level steering fare poorly: concept bottleneck steering corrects 20% of missed cases while disrupting 53% of correct ones, indistinguishable from random perturbation. Liu [2] explain the mechanism: the failure-mode direction overlaps 85–88% with task-critical computation in the residual stream, making representation-space correction structurally self-defeating.

We report a logit-level approach that sidesteps this collision. A constant bias toward hedge tokens—applied to the output distribution, not the representation space—produces a training-free phase transition from confident fabrication to appropriate epistemic hedging. The mechanism is not steering but *denoising*: the model’s uncertainty signal is always present but faint;

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the logit bias amplifies it past the fabrication-confidence noise floor. At bias = 5.0, the entropy at the decision token increases $3.2\times$ over baseline, and the model that was confidently inventing the Grenvillia Trench now correctly identifies it as unknown.

The finding has three implications developed in this paper. First, confabulation is not one pathology: fabrication and imprecision respond to different intervention levels and may require different correction channels (Section 4). Second, the dose required for correction correlates with the strength of the model’s knowledge anchoring, suggesting that geometric detection magnitude can directly prescribe the logit-bias dose—closing the diagnostic-to-treatment loop (Section 7). Third, the denoising principle—amplifying a faint signal that is already present rather than imposing an external direction—connects logit-level confabulation correction to cache-level behavioral steering and geometric presence detection as three instances of one mechanism (Section 5).

2 Related Work

The detection–correction gap. Basu et al. [1] and Liu [2] document the gap from complementary angles: near-perfect internal detection of failure modes coexists with structurally limited correction via representation-level steering. Our value-only correction framework [4] operates in a different subspace (values rather than the residual stream), partially sidestepping the overlap; the present work avoids the representation space entirely.

Confabulation detection and correction. Prior work from this lab achieves within-model confabulation detection at AUROC ≥ 0.969 via KV-cache spectral features. The E-matrix prescription (companion paper) maps pathology-specific correction vectors in emotion-geometric space, with hostile-valence correcting confabulation ($d = -1.534$) and desperate-valence correcting deception ($d = 1.286$). The present work complements cache-level correction with a logit-level channel for the imprecision subtype.

Logit-level interventions. Logit bias is a standard inference-time parameter but has received limited systematic study as a confabulation intervention. Entropy-based gating (applying bias only above an entropy threshold) has been proposed but, as we show, performs worse than constant application due to generation-coherence disruption.

3 Methods

[CC’s section — methods, experimental design, classification.]

4 Results: Two Confab Subtypes and the Dose–Response Curve

[CC’s section — results, taxonomy, dose-response.]

5 The Denoising Principle

Three apparently different problems share a mechanism: in each case, the corrective information is already present in the model, and the intervention is amplification rather than creation.

Confabulation correction (this paper). The model’s hedge signal exists in the entropy profile at the decision token, measurably nonzero even when the output is confidently fabricated. In the powered study, baseline entropy at token 30 is 0.24; at bias = 5.0 it reaches 0.77—a $3.2\times$ amplification. The transition is not gradual: the model confabulates at bias ≤ 3.0 (entropy

≤ 0.45) and hedges at bias = 5.0 (entropy = 0.77). The logit bias is a gain knob on the faint uncertainty signal; below threshold, fabrication-confidence noise drowns it; above threshold, uncertainty surfaces and behavior flips.

Presence detection (companion paper). A trained persona’s geometric signature in the value space is faint—detectable at family-wise $p = 0.0005$ on public character-trained models [3], but low-rank and small relative to the architectural mode. Removing the dominant singular component (skip-SV1) is a gain operation: it suppresses the loud architectural prior so the quiet trained signal becomes visible. The principle is the same: amplify the faint signal by reducing the noise, not by creating the signal.

Cache-level correction (E-matrix). The Oracle Loop detects confabulation and deception by their geometric signatures in the KV cache. The E-matrix correction injects a contrastive value-delta along the detected pathology axis—gain applied to the pathology signal. The 13-saves / 12-hurts overcorrection at $n = 200$ is what happens when the gain is too high: the amplified signal overshoots the target basin.

In each case, the intervention surfaces what is already present. We call this *denoising* rather than *steering*: steering implies imposing an external direction; denoising implies surfacing an internal one. The distinction matters because denoising has a natural self-limiting property (once the signal exceeds the noise floor, further amplification is redundant), while steering does not.

6 Why Logit-Level Correction Sidesteps the Gap

The detection–correction gap [1] arises because the failure-mode direction and task-critical computation share 85–88% of the representation space [2]. Correcting along the failure-mode direction in that space necessarily disrupts the task.

Logit-level bias operates on a different surface entirely. It does not modify the model’s internal representations; it modifies the *output distribution*—the probability mass assigned to hedge tokens versus continuation tokens at each generation step. The model’s internal computation proceeds unperturbed; only the final selection among computed options is reweighted. This is why it avoids the 20%-corrected / 53%-disrupted ratio: the task-critical representations are untouched.

The cost is precision. Logit-level bias is a blunt instrument—it biases *all* generation toward hedging, not just the uncertain parts. But the powered study shows this cost is manageable: legitimate Fermi estimation is unperturbed at all bias levels (0.0 through 5.0), with only mild hedging language appearing before correct estimates at the highest dose. The intervention is selective because the hedge tokens it promotes are already near-threshold for uncertain content but far below threshold for confident factual content.

7 Detection as Prescription

The most striking implication of the dose–response data is that the Oracle Loop’s geometric detection may directly prescribe the logit-bias dose.

In the pilot study (7B ablated, 3 fictional entities), entities with strong knowledge anchoring—the fictional “Kaminski Strait” mapping onto the real Bering Strait—resist correction until bias = 5.0. Entities with weak anchoring—the “Treaty of Bordenholm”—correct at bias = 1.0. The per-entity threshold correlates with how strongly the fabrication anchors to real knowledge in the model’s representation space.

If the Oracle Loop’s confab detection magnitude (`confab_proj`) tracks this anchoring strength, then the diagnostic signal directly prescribes the minimum effective dose: strong detection $\rightarrow$ high bias, weak detection $\rightarrow$ low bias. This would constitute a self-calibrating correction loop: measure the fabrication strength geometrically, set the amplification accordingly, and the model corrects itself at the minimum effective dose.

The powered study (175 trials, 20 fictional entities) provides the dose–response curve; correlating `confab_proj` magnitude with per-entity threshold dose is the validation step. This analysis requires running the Oracle Loop’s geometric detector on the powered-study prompts, which is planned but not yet complete.

8 Limitations

The logit-bias approach has four principal limitations. First, the powered study uses a single abliterated model (Qwen3.5-27B); cross-architecture replication is needed. Second, the behavioral classification in the powered study uses a simple pattern-based classifier; an LLM-judge pass is required for precise confabulation rates (the entropy signal is the more reliable metric). Third, the detection-as-prescription bridge (`confab_proj` $\rightarrow$ dose) is proposed but not yet validated—the geometric detector has not been run on the powered-study entities. Fourth, the relationship between logit-level and cache-level correction channels (whether they compose, interfere, or are complementary for different subtypes) is theoretical, not empirical.

9 Conclusion

The model knows what it does not know. The uncertainty signal is present at the decision token, faint but real. A constant logit bias amplifies it past the fabrication-confidence noise floor, producing a training-free phase transition from confident fabrication to honest hedging. The intervention works because it operates on the output distribution, sidestepping the representation-space collision that defeats steering approaches. And the dose required for correction tracks the strength of the model’s knowledge anchoring, opening the possibility that geometric detection can directly prescribe the minimum effective intervention.

The principle is denoising, not steering: surface what is already there, at the minimum gain that lets the signal through. The same principle governs presence detection and cache-level correction. In each case, the model contains the answer to its own pathology. The intervention is the gain.

References

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